The TV survey gives E(X) = 1.9. Which interpretation is correct?

TVs in household	0	1	2	3	4
Households	5	30	40	20	5

A. The surveyed households averaged about 1.9 TVs.

B. Most of the households have exactly 1.9 TVs.

C. Every household surveyed has 1 or 2 TVs.

D. There is a 1.9% chance of owning a TV.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-M-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q4/5

A store recorded complaints for 50 days (table). Find the expected number of complaints in a day.

Complaints in a day	0	1	2	3
Days	20	15	10	5

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q5/5

SET 49 • HSS-MD.A.4

BACKWARD 1

Find relative frequencies • Grade 7 • Statistics & Probability • 7.SP.C.6

In 100 households, 40 have 2 TVs. What is the relative frequency? Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 49 • BACKWARD 1 • Q1/5

On 20 of 50 days there were no complaints. What is the relative frequency? Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 49 • BACKWARD 1 • Q2/5

True or false?

Relative frequencies of all categories add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 49 • BACKWARD 1 • Q3/5

On 5 of 50 days there were 3 complaints. Relative frequency?

A. 5

B. 0.1

C. 0.05

D. 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 49 • BACKWARD 1 • Q4/5

True or false?

A relative frequency of 0.25 means 25 observations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • HSS-MD.A.4

BACKWARD 2

Compute an expected value • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

X takes 0, 1, 2 with probabilities 0.5, 0.3, 0.2. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 49 • BACKWARD 2 • Q1/5

True or false?

$$E(X) = \Sigma\, x \cdot P(X = x).$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 49 • BACKWARD 2 • Q2/5

X takes 1 and 3 with probabilities 0.5 each. E(X)?

A. 4

B. 2

C. 1.5

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 49 • BACKWARD 2 • Q3/5

X takes 0, 2, 4 with probabilities 0.25, 0.5, 0.25. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 49 • BACKWARD 2 • Q4/5

True or false?

E(X) is found by adding the probabilities.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • HSS-MD.A.4

FORWARD 1

Compare empirical and theoretical results • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.4

A die was rolled 60 times: the faces 1-6 appeared 7, 13, 9, 11, 14, 6 times. Find the empirical P(6) and the theoretical P(6) for a fair die. Give decimals rounded to two places.

Empirical: _____

Theoretical: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • FORWARD 1 • Q1/5

Using the 60 rolls (faces 1-6 appeared 7, 13, 9, 11, 14, 6 times), find the empirical expected value of a roll.

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • FORWARD 1 • Q2/5

True or false?

For repeated independent rolls of a fair die, the empirical probabilities tend to get closer to 1/6 as the number of rolls grows.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • FORWARD 1 • Q3/5

Which result would most suggest the die is unfair?

A. In 600 rolls, a 6 appeared 104 times.

B. In 6,000 rolls, a 6 appeared 1,800 times.

C. In 60 rolls, a 6 appeared 6 times.

D. In 6 rolls, no 6 appeared.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • FORWARD 1 • Q4/5

True or false?

An empirical distribution from 10 observations is as reliable as one from 10,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • FORWARD 1 • Q5/5

SET 49 • HSS-MD.A.4

FORWARD 2

Find the expected payoff of a game of chance • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.a

A $1 ticket wins $200 with probability 1/500 (and nothing otherwise). Find the expected net gain per ticket.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 49 • FORWARD 2 • Q1/5

True or false?

A game with negative expected net gain loses money for the player in the long run.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 49 • FORWARD 2 • Q2/5

A spinner: win $6 with probability 1/3, lose $3 otherwise. Find the expected gain.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 49 • FORWARD 2 • Q3/5

A game's expected net gain is −$0.60. Which interpretation is correct?

A. The player wins $0.60 on about half the plays.

B. Every player loses exactly $0.60 on each play.

C. The game is fair, since some plays win.

D. Over many plays, a player loses about $0.60 per play.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 49 • FORWARD 2 • Q4/5

True or false?

A game with expected net gain $0 is called fair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 49 • FORWARD 2 • Q5/5

Weigh the payoffs of a game of chance (+)

MAIN	Probability & Statistics main question	**HSS-MD.B.5.a**
BACKWARD 1	Compute an expected value	**HSS-MD.A.2**
BACKWARD 2	Add and subtract signed numbers	**7.NS.A.3**
FORWARD 1	Set a price that makes a game fair	**HSS-MD.B.5.a**
FORWARD 2	Compare decisions by expected value	**HSS-MD.B.5.b**

SET 50 • HSS-MD.B.5.a

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.a

A lottery ticket costs $2 and wins $500 with probability 1/1000; otherwise it wins nothing. Find the expected net gain per ticket.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • MAIN • Q1/5

A game: roll a fair die. If it shows 6, you win $5; otherwise you lose $1. Find the expected gain per play.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • MAIN • Q2/5

A raffle sells 500 tickets at $5 each for one $1,000 prize. Find the expected net gain per ticket.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • MAIN • Q3/5

A lottery ticket has expected net gain $-\$1.50$. Which interpretation is correct?

A. The lottery is fair, because some tickets win a prize.

B. Every ticket loses exactly $1.50 for the person who buys it.

C. Over many tickets, a buyer loses about $1.50 per ticket on average.

D. Buyers win $1.50 on about half of the tickets they buy.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-M-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • MAIN • Q4/5

A spinner wins $8 with probability 1/4 and loses $2 otherwise. Find the expected gain per spin.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • MAIN • Q5/5

SET 50 • HSS-MD.B.5.a

BACKWARD 1

Compute an expected value • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

X = 10 with probability 0.1 and X = 0 otherwise. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 50 • BACKWARD 1 • Q1/5

True or false?

If X = 10 with probability 0.1 and 0 otherwise, then E(X) = 10(0.1) + 0(0.9).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 50 • BACKWARD 1 • Q2/5

$X = 4$ with probability 0.25 and $X = -1$ otherwise. $E(X)$?

A. 1

B. 0.25

C. -0.75

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 50 • BACKWARD 1 • Q3/5

X = 100 with probability 0.01 and 0 otherwise. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 50 • BACKWARD 1 • Q4/5

True or false?

$E(X) = 5$ when $X = 10$ with probability 0.2 and $X = 0$ otherwise.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • HSS-MD.B.5.a

BACKWARD 2

Add and subtract signed numbers • Grade 7 • The Number System • 7.NS.A.3

Compute $0.5 - 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 50 • BACKWARD 2 • Q1/5

Compute 5(1/6) − 1(5/6).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 50 • BACKWARD 2 • Q2/5

True or false?

$2 - 1.5 = 0.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 50 • BACKWARD 2 • Q3/5

Compute $2 - 5$.

A. 3

B. -7

C. 7

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 50 • BACKWARD 2 • Q4/5

True or false?

$-1.5 + 2 = -3.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 50 • BACKWARD 2 • Q5/5

SET 50 • HSS-MD.B.5.a

FORWARD 1

Set a price that makes a game fair • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.a

A game pays $12 if a fair die shows 6 and nothing otherwise. What price makes the game fair?

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • FORWARD 1 • Q1/5

A raffle has one $1,000 prize and 400 tickets. What ticket price makes the raffle fair?

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • FORWARD 1 • Q2/5

True or false?

A game is fair when the price equals the expected payout.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • FORWARD 1 • Q3/5

A game costs $3 and pays $10 with probability 0.25 (and nothing otherwise). What is the expected net gain over 100 plays?

A. −$0.50

B. $50

C. $250

D. −$50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions

HSS-MD.B.5.a • Set 50 • FORWARD 1 • Q4/5

True or false?

A raffle with 400 tickets and a $1,000 prize is fair at $5 per ticket.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • FORWARD 1 • Q5/5

SET 50 • HSS-MD.B.5.a

FORWARD 2

Compare decisions by expected value • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.b

Option X gains $1,000 with probability 0.6 and loses $500 with probability 0.4. Option Y gains $300 for sure. Find the expected gain of each.

X: _____

Y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 50 • FORWARD 2 • Q1/5

Option X: expected gain $400 but a 40% chance of losing $500. Option Y: $300 for sure. Which statement is correct?

A. X is certain to gain $400 for whoever chooses it.

B. X has the higher expected value, but Y has no risk of loss.

C. X and Y have exactly the same expected value.

D. Y has the higher expected value of the two.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 50 • FORWARD 2 • Q2/5

True or false?

Expected value describes the long-run average, not what happens in one decision.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 50 • FORWARD 2 • Q3/5

Insurance costs $200. Without it, a $1,500 repair is needed with probability 0.1 (otherwise no repair is needed). What is the expected repair cost without insurance?

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 50 • FORWARD 2 • Q4/5

True or false?

Buying insurance always has a greater expected value than not buying it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 50 • FORWARD 2 • Q5/5

Evaluate and compare strategies by expected value (+)

MAIN	Probability & Statistics main question	**HSS-MD.B.5.b**
BACKWARD 1	Find the expected payoff of one choice	**HSS-MD.B.5.a**
BACKWARD 2	Read a distribution and its mean	**HSS-MD.A.2**
FORWARD 1	Find break-even probabilities	**HSS-MD.B.5.b**
FORWARD 2	Make fair decisions with probability	**HSS-MD.B.6**

SET 51 • HSS-MD.B.5.b

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.b

Plan A: premium $1,200, deductible $500. Plan B: premium $900, deductible $1,000. In a year, one claim larger than $1,000 occurs with probability 0.2, and otherwise there is no claim. When the claim occurs, the owner pays the plan's deductible. Find the expected yearly cost of each plan (premium plus expected deductible).

Plan A: _____

Plan B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • MAIN • Q1/5

Plan A has expected cost $1,300 and plan B $1,100 per year. Which plan is better on the basis of expected cost?

A. Plan B

B. It cannot be decided

C. Plan A

D. They are equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • MAIN • Q2/5

Option X gains $2,000 with probability 0.3 and loses $400 with probability 0.7. Option Y gains $250 for sure. Find the expected gain of each.

X: _____

Y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • MAIN • Q3/5

X has expected gain $320 with a 70% chance of losing $400; Y gains $250 for sure. Which statement is correct?

A. Y is better on expected value, so everyone should prefer Y.

B. X is certain to gain $320 for the person who chooses it.

C. X and Y have the same expected value, so they are equal.

D. X is better on average, but someone who cannot risk a loss may prefer Y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-M-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • MAIN • Q4/5

A phone warranty costs $150. Without it, a repair costing $800 is needed with probability 0.1 (otherwise no repair is needed). What is the expected repair cost without the warranty, and how much more is the warranty than that?

Expected cost: _____

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • MAIN • Q5/5

SET 51 • HSS-MD.B.5.b

BACKWARD 1

Find the expected payoff of one choice • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.a

A choice gains $200 with probability 0.5 and $0 otherwise. Find the expected gain.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 51 • BACKWARD 1 • Q1/5

A choice loses $1,000 with probability 0.2 and $0 otherwise. Find the expected loss.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 51 • BACKWARD 1 • Q2/5

True or false?

A sure gain of $300 has expected value $300.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 51 • BACKWARD 1 • Q3/5

A bet wins $50 with probability 0.3 and loses $20 otherwise. Expected gain?

A. $30

B. −$14

C. $15

D. $1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 51 • BACKWARD 1 • Q4/5

True or false?

A choice with expected gain $0 can never make money.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 51 • BACKWARD 1 • Q5/5

SET 51 • HSS-MD.B.5.b

BACKWARD 2

Read a distribution and its mean • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

True or false?

The option with the larger expected value is better on average.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 51 • BACKWARD 2 • Q1/5

$X = 1000$ with probability 0.6 and -500 with probability 0.4. Find $E(X)$.

$E(X)$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 51 • BACKWARD 2 • Q2/5

Which has the larger expected value: $50 for sure, or $120 with probability 0.5 (and $0 otherwise)?

A. cannot tell

B. they are equal

C. $120 with probability 0.5

D. $50 for sure

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 51 • BACKWARD 2 • Q3/5

Cost = 500 with probability 0.2 and 0 otherwise. Find the expected cost.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 51 • BACKWARD 2 • Q4/5

True or false?

Expected cost = cost × (1 − probability).

A. True

B. False

SET 51 • HSS-MD.B.5.b

FORWARD 1

Find break-even probabilities • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.b

A $150 warranty covers an $800 repair. For what repair probability is the expected repair cost equal to $150? Give a decimal.

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • FORWARD 1 • Q1/5

True or false?

If the repair probability is below the break-even value, skipping the warranty has the lower expected cost.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • FORWARD 1 • Q2/5

Plan A costs $1,200 plus a $500 deductible; plan B costs $900 plus a $1,000 deductible. For what claim probability p are the expected costs equal?

p: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • FORWARD 1 • Q3/5

With claim probability 0.8, which plan has the lower expected cost (A: 1200 + 500p; B: 900 + 1000p)?

A. Plan A

B. Plan B

C. They are equal

D. Neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F1-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • FORWARD 1 • Q4/5

True or false?

The better plan never depends on the claim probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • FORWARD 1 • Q5/5

SET 51 • HSS-MD.B.5.b

FORWARD 2

Make fair decisions with probability • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

Four friends want to decide fairly with one fair die who gets a ticket. Which method is fair?

A. Assign 1-4 to the friends; on a 5 or 6, roll again.

B. Assign 1-4 and give 5 and 6 to the first friend.

C. Give the ticket to whoever guesses the roll.

D. Give 1-3 to one friend and 4-6 to the other three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 51 • FORWARD 2 • Q1/5

True or false?

A method is fair if every person has the same probability of being chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 51 • FORWARD 2 • Q2/5

Ana has 4 tickets, Ben 6 and Cy 10 in a drawing of 20 tickets; one ticket is drawn, each equally likely. What is P(Ana wins)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 51 • FORWARD 2 • Q3/5

In a game with two fair dice rolled independently, A wins if the sum is 7 or less; otherwise B wins. Is it fair?

A. No: A wins with probability 7/12.

B. No: A wins with probability 6/11.

C. Yes: A has 6 sums and B has 5.

D. Yes: each player has 1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 51 • FORWARD 2 • Q4/5

True or false?

Letting the oldest person choose is a fair random method.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 51 • FORWARD 2 • Q5/5

Use probabilities to make fair decisions (+)

MAIN	Probability & Statistics main question	**HSS-MD.B.6**
BACKWARD 1	Simulate with random digits	**HSS-IC.A.2**
BACKWARD 2	Find probabilities in a uniform model	**7.SP.C.7.a**
FORWARD 1	Design fair methods for awkward numbers	**HSS-MD.B.6**
FORWARD 2	Analyze a decision with a medical test	**HSS-MD.B.7**

SET 52 • HSS-MD.B.6

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

Three friends want to decide fairly who gets the last ticket. Which method is fair?

A. Roll a die: 1 Ana, 2 Ben, 3-6 Cy.

B. Roll a die: 1-2 Ana, 3-4 Ben, 5-6 Cy.

C. Flip a coin: heads Ana, tails Ben; Cy only on an edge.

D. Ana chooses, since she asked first.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • MAIN • Q1/5

One of four people, A, B, C and D, is chosen with random digits: 0-1 = A, 2-3 = B, 4-5 = C, 6-7 = D, and 8-9 are skipped. Reading line 2 of the table from the left, who is chosen? Enter A, B, C or D.

Line	Digits	Line	Digits
1	15881 39986 98797 30911	5	17784 86488 76634 40029
2	41707 53645 56815 18844	6	78753 47787 68045 71784
3	72940 55761 69871 68976	7	72671 47513 62701 07981
4	84697 88093 20819 67273	8	93319 08452 05997 28379

Person: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • MAIN • Q2/5

Two players each flip a fair coin, independently. Ana wins if the coins match; Ben wins if they differ. What is P(Ana wins)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • MAIN • Q3/5

Two fair dice are rolled independently. Player A wins if the sum is 6, 7 or 8; otherwise B wins. Is the game fair?

A. Yes: the sums are split fairly.

B. Yes: each player wins with probability 1/2.

C. No: A wins with probability 3/11.

D. No: A wins with probability 4/9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-M-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • MAIN • Q4/5

In a drawing, Ana has 3 tickets, Ben 5 and Cy 2, out of 10; one ticket is drawn, each equally likely. What is P(Ben wins)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • MAIN • Q5/5

SET 52 • HSS-MD.B.6

BACKWARD 1

Simulate with random digits • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

True or false?

Each digit 0-9 in a random-digit table is equally likely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 52 • BACKWARD 1 • Q1/5

To choose among 4 people with single digits, which assignment is fair?

A. 0-2, 3-5, 6-8 and 9

B. 0-1, 2-3, 4-5, 6-9

C. 0, 1, 2, 3 and stop

D. 0-1, 2-3, 4-5, 6-7 and skip 8-9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-B1-Q2**

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 52 • BACKWARD 1 • Q2/5

Using 0-1 = A, 2-3 = B, 4-5 = C, 6-7 = D, skip 8-9, what does the digit 9 mean?

Answer (enter skip or a letter): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 52 • BACKWARD 1 • Q3/5

True or false?

Skipping digits 8 and 9 makes the assignment unfair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 52 • BACKWARD 1 • Q4/5

One of four people is chosen with random digits: 0-1 = A, 2-3 = B, 4-5 = C, 6-7 = D, and 8-9 are skipped (read the next digit). The digits read 8, 9, 3. Who is chosen?

A. B

B. A

C. no one

D. D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 52 • BACKWARD 1 • Q5/5

SET 52 • HSS-MD.B.6

BACKWARD 2

Find probabilities in a uniform model • Grade 7 • Statistics & Probability • 7.SP.C.7.a

A fair six-sided die is rolled. What is P(an odd number)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 52 • BACKWARD 2 • Q1/5

True or false?

With a fair coin, heads and tails each have probability 1/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 52 • BACKWARD 2 • Q2/5

Names of 5 people are in a hat. P(a particular person is drawn)?

A. 1/2

B. 1/4

C. 5

D. 1/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 52 • BACKWARD 2 • Q3/5

Two fair coins are flipped independently. What is P(they match)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 52 • BACKWARD 2 • Q4/5

True or false?

A spinner with one half red and two quarters blue and green is fair among the three colors.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 52 • BACKWARD 2 • Q5/5

SET 52 • HSS-MD.B.6

FORWARD 1

Design fair methods for awkward numbers • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

Five people must be chosen from fairly with one fair die. Which method works?

A. Assign 1-5 to the people and give 6 to the first person.

B. Assign 1-5 to the people; on a 6, roll again.

C. Assign 1-6 and let the sixth face mean "everyone".

D. Roll and divide the result by 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F1-Q1**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • FORWARD 1 • Q1/5

Five people are assigned the faces 1-5 of a fair die; on a 6 the die is rolled again, until a 1-5 comes up. What is each person's probability of being chosen? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • FORWARD 1 • Q2/5

True or false?

A method that uses one roll of a fair die and gives one person 2 of the 6 faces and each other person 1 face is unfair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • FORWARD 1 • Q3/5

One of 3 people is chosen with random two-digit numbers 00-99. Each person gets the same number of values, as many of the 100 numbers as possible are used, and a skipped number means reading the next number. How many numbers does each person get?

Numbers each: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • FORWARD 1 • Q4/5

True or false?

Drawing names one at a time from a hat without replacement is unfair to the last name.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • FORWARD 1 • Q5/5

SET 52 • HSS-MD.B.6

FORWARD 2

Analyze a decision with a medical test • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

Of 2,000 people, 1% have a disease. How many have it?

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 52 • FORWARD 2 • Q1/5

A test is positive for 95% of the 20 people with a disease. How many of them test positive?

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 52 • FORWARD 2 • Q2/5

True or false?

A test can give positive results for people who do not have the disease.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 52 • FORWARD 2 • Q3/5

Of 1,980 healthy people, 10% test positive. How many false positives are there?

A. 10

B. 198

C. 19

D. 1,782

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 52 • FORWARD 2 • Q4/5

True or false?

If a test is positive for 95% of sick people, then 95% of positives are sick.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 52 • FORWARD 2 • Q5/5

Analyze decisions about medical testing (+)

MAIN	Probability & Statistics main question	**HSS-MD.B.7**
BACKWARD 1	Read a two-way table as a sample space	**HSS-CP.A.4**
BACKWARD 2	Find a conditional probability as a fraction of outcomes	**HSS-CP.B.6**
FORWARD 1	See how prevalence changes the answer	**HSS-MD.B.7**
FORWARD 2	Analyze strategies in games and quality control	**HSS-MD.B.7**

SET 53 • HSS-MD.B.7

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

1,000 people are tested. 2% have a disease; the test is positive for 90% of people with the disease. How many have the disease, and how many of them test positive?

Have the disease: _____

Test positive: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • MAIN • Q1/5

Of the 980 people without the disease, 5% test positive. How many false positives are there?

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • MAIN • Q2/5

Use the table for 1,000 people. A person tests positive. What is P(has disease given positive)? Give a fraction in lowest terms.

	Tests positive	Tests negative	Total
Has disease	18	2	20
Does not	49	931	980
Total	67	933	1000

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • MAIN • Q3/5

In the table, P(disease given positive) = 18/67, about 27%. What does this show?

	Tests positive	Tests negative	Total
Has disease	18	2	20
Does not	49	931	980
Total	67	933	1000

A. Most people who test positive do not have the disease.

B. Most people who have the disease test negative.

C. The test is wrong 73% of the time for everyone.

D. A positive test means a 90% chance of disease.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-M-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • MAIN • Q4/5

Use the table. A person tests negative. What is P(does not have the disease given negative)? Give a fraction in lowest terms.

	Tests positive	Tests negative	Total
Has disease	18	2	20
Does not	49	931	980
Total	67	933	1000

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • MAIN • Q5/5

SET 53 • HSS-MD.B.7

BACKWARD 1

Read a two-way table as a sample space • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

In the test table, how many people test positive in all?

	Tests positive	Tests negative	Total
Has disease	18	2	20
Does not	49	931	980
Total	67	933	1000

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 53 • BACKWARD 1 • Q1/5

In the test table, how many people test negative in all?

	Tests positive	Tests negative	Total
Has disease	18	2	20
Does not	49	931	980
Total	67	933	1000

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 53 • BACKWARD 1 • Q2/5

True or false?

In the test table, 20 people have the disease.

	Tests positive	Tests negative	Total
Has disease	18	2	20
Does not	49	931	980
Total	67	933	1000

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-B1-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 53 • BACKWARD 1 • Q3/5

In the test table, which cell counts false positives?

	Tests positive	Tests negative	Total
Has disease	18	2	20
Does not	49	931	980
Total	67	933	1000

A. does not have the disease, tests positive

B. does not have the disease, tests negative

C. has the disease, tests negative

D. has the disease, tests positive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 53 • BACKWARD 1 • Q4/5

True or false?

In the test table, false negatives number 18.

	Tests positive	Tests negative	Total
Has disease	18	2	20
Does not	49	931	980
Total	67	933	1000

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 53 • BACKWARD 1 • Q5/5

SET 53 • HSS-MD.B.7

BACKWARD 2

Find a conditional probability as a fraction of outcomes • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6

Of 67 people who test positive, 18 have the disease. What fraction is that?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 53 • BACKWARD 2 • Q1/5

True or false?

P(disease given positive) uses the positive column as the sample space.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 53 • BACKWARD 2 • Q2/5

Of 20 people with the disease, 18 test positive. P(positive given disease)?

A. 18/67

B. 9/10

C. 2/20

D. 1/50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-B2-Q3**

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 53 • BACKWARD 2 • Q3/5

Of 980 healthy people, 49 test positive. What is P(positive given healthy)? Give a decimal.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 53 • BACKWARD 2 • Q4/5

True or false?

P(positive given disease) = P(disease given positive).

A. True

B. False

SET 53 • HSS-MD.B.7

FORWARD 1

See how prevalence changes the answer • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

In a population of 1,000 people, 10% have a disease. A test is positive for 90% of the people with the disease and for 5% of the people without it. How many true positives and how many false positives are there?

True positives: _____

False positives: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 1 • Q1/5

With 90 true positives and 45 false positives, what is P(disease given positive)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 1 • Q2/5

True or false?

The same test gives a higher P(disease given positive) when the disease is more common.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 1 • Q3/5

Why do doctors often order a second test after a positive result for a rare disease?

A. Negative results are never correct.

B. The first test is always wrong.

C. A second test lowers the prevalence.

D. Many first positives are false positives.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 1 • Q4/5

True or false?

For a rare disease, a positive result makes having the disease certain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 1 • Q5/5

SET 53 • HSS-MD.B.7

FORWARD 2

Analyze strategies in games and quality control • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

A box of 10 parts has 2 defective. An inspector checks 2 parts chosen at random, without replacement (every pair equally likely). What is P(no defective part is found)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 2 • Q1/5

True or false?

Checking more parts raises the chance of finding a defective part.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 2 • Q2/5

A test question has 4 choices and one correct answer; a correct answer earns 1 point, a wrong answer loses 1/3 point, and a blank earns 0. A guess picks at random among the choices still in play. What is the expected score of a guess with no choices eliminated, and of a guess after one wrong choice is eliminated? Give a fraction in lowest terms.

Random guess: _____

After eliminating one: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 2 • Q3/5

A test question has 4 choices and one correct answer: +1 point if correct, −1/3 point if wrong, 0 if left blank. A guess picks at random among the choices still in play. When does a guess have a positive expected score?

A. always, even with no information

B. after eliminating at least one wrong choice

C. only on the last question of the test

D. never, under any circumstances

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions

HSS-MD.B.7 • Set 53 • FORWARD 2 • Q4/5

True or false?

A strategy with a higher expected value always wins in a single try.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 2 • Q5/5

Analyze strategies and decisions with probability (+)

MAIN	Probability & Statistics main question	**HSS-MD.B.7**
BACKWARD 1	Compare expected values of choices	**HSS-MD.A.2**
BACKWARD 2	Count selections to find probabilities	**HSS-CP.B.9**
FORWARD 1	Weigh probabilities against consequences	**HSS-MD.B.7**
FORWARD 2	Combine several probability tools in one decision	**HSS-MD.B.7**

SET 54 • HSS-MD.B.7

MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

A hockey team trails by 1 goal with 2 minutes left, and its only goal is to avoid losing, which requires tying. Keeping the goalie gives a 0.08 chance of tying; pulling the goalie for an extra skater gives a 0.20 chance (allowing another goal still means a loss, as before). By how many percentage points does pulling the goalie raise the chance of tying, and how many times as large is that chance?

Percentage points: _____

Times as large: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • MAIN • Q1/5

The team loses unless it ties. Keeping the goalie gives a 0.08 chance of tying; pulling gives 0.20. Which strategy is better?

A. Pull the goalie: it gives the better chance of avoiding a loss.

B. Keep the goalie: allowing another goal would be worse.

C. Keep the goalie: 0.08 is the safer probability.

D. Either one: both chances are less than one-half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-M-Q2**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • MAIN • Q2/5

A box of 12 parts has 3 defective. An inspector checks 2 parts chosen at random, without replacement (every pair equally likely). What is P(no defective part is found)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • MAIN • Q3/5

A test question has 5 choices and one correct answer; a correct answer earns 1 point, a wrong answer loses 1/4 point, and a blank earns 0. A guess picks at random among the choices still in play. Find the expected score of a guess with no choices eliminated, and of a guess after eliminating one wrong choice. Give a fraction in lowest terms.

Random guess: _____

After eliminating one: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • MAIN • Q4/5

A company can test 2 or 5 parts from each box of 10 (2 defective). With 2 tested, P(missing both defectives) = 28/45; with 5 tested, it is 2/9. Which is correct?

A. Testing 5 parts lowers the chance of missing the defects.

B. Testing 5 parts always finds both defects.

C. Testing 2 parts finds the defects more often.

D. The number tested does not change anything.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-M-Q5**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • MAIN • Q5/5

SET 54 • HSS-MD.B.7

BACKWARD 1

Compare expected values of choices • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

Strategy A scores 3 with probability 0.5 and 0 otherwise. Find its expected score.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 54 • BACKWARD 1 • Q1/5

True or false?

Between two strategies, the one with the larger expected value is better on average.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 54 • BACKWARD 1 • Q2/5

Strategy B scores 2 for sure; strategy A scores 3 with probability 0.5 and 0 otherwise. Which is better on average?

A. cannot tell

B. A

C. B

D. equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-B1-Q3**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 54 • BACKWARD 1 • Q3/5

A guess earns 1 with probability 1/3 and loses 1/3 otherwise. Expected score? Give a fraction in lowest terms.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 54 • BACKWARD 1 • Q4/5

True or false?

An expected score of 0 means the strategy always scores 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 54 • BACKWARD 1 • Q5/5

SET 54 • HSS-MD.B.7

BACKWARD 2

Count selections to find probabilities • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

Compute 10C2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 54 • BACKWARD 2 • Q1/5

Compute 8C2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 54 • BACKWARD 2 • Q2/5

True or false?

$8C5 = 56$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 54 • BACKWARD 2 • Q3/5

Compute 10C5.

A. 30240

B. 120

C. 252

D. 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 54 • BACKWARD 2 • Q4/5

True or false?

P(no defectives in 2 of 10 with 2 defective) = 8/10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • HSS-MD.B.7

FORWARD 1

Weigh probabilities against consequences • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

A product test costs $50 per box and finds the defects with probability 0.8. An undetected defective box costs $400. For a defective box, what is the expected cost of shipping without testing, and with testing (test cost plus expected cost of a missed defect)?

Without testing: _____

With testing: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 1 • Q1/5

True or false?

A strategy can have a better expected value even if it sometimes produces the worst outcome.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 1 • Q2/5

On a test, each question has 4 choices and one correct answer: +1 point if correct, −1/3 point if wrong, 0 if blank. A guess picks at random among the choices still in play. Which statement about strategy is correct?

A. A blind guess averages −1/12, less than a blank, so never guess without eliminating.

B. A blind guess and a blank both average 0; a guess after one elimination averages 1/9.

C. Eliminating one wrong choice leaves the average score of a guess at 0.

D. A blind guess averages 1/4, more than a blank, so guess on every question.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F1-Q3**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 1 • Q3/5

A question has 4 choices and one correct answer: +1 point if correct, −1/3 point if wrong, 0 if blank. Two wrong choices are eliminated, and a guess picks at random between the 2 left. What is the expected score of the guess? Give a fraction in lowest terms.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 1 • Q4/5

True or false?

Probability analysis can tell exactly what will happen in one game.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 1 • Q5/5

SET 54 • HSS-MD.B.7

FORWARD 2

Combine several probability tools in one decision • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

A quiz has 5 questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others), and a student guesses on all of them. What is the expected number correct, and P(all wrong)? Give each answer as a fraction in lowest terms.

Expected correct: _____

P(all wrong): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 2 • Q1/5

True or false?

P(at least one correct) = 1 − P(all wrong).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 2 • Q2/5

A quiz has 5 questions (each question has 4 choices and one correct answer; each guess picks a choice at random, independently of the others), and a student guesses on all of them. What is P(at least one correct)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 2 • Q3/5

A team must choose between a play that gains 2 points with probability 0.5 (and 0 otherwise) and one that gains 3 points with probability 0.3 (and 0 otherwise). Which has the higher expected gain?

A. equal

B. cannot tell

C. the 2-point play

D. the 3-point play

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q4**

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 2 • Q4/5

True or false?

On one final play, option P scores 2 points with probability 0.5 (otherwise 0) and option Q scores 3 points with probability 0.3 (otherwise 0). The team needs at least 3 points on this play to win. Option P is the better choice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 2 • Q5/5